Sterility of triangular pole-hiding over lines and hyperbolas in $\mathbb{C}^{2*}$

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Abstract

The Jacobian conjecture fails in dimension $n \geq 3$: the July 2026 counterexample of Alpöge–Fable is a Keller map of $\mathbb{C}^3$ built from a rational coordinate whose poles are hidden on a hyperplane and cancelled by a ramified composition. We study whether this mechanism — *triangular pole hiding* — can operate in the plane, where the conjecture remains open. We formalize the mechanism as a *scaffold* over a pole curve $\Gamma \subset \mathbb{C}^2$ and prove two sterility theorems. Over the hyperbola $\Gamma = \{xy = 1\}$ no scaffold exists at all. Over the line $\Gamma = \{x = 0\}$, every scaffold in every stratum of an explicit, machine-verified partition of the configuration space save one named sliver is an affine triangular automorphism; a scaffold outside the tame side would *be* a counterexample to the two-dimensional conjecture. In the remaining sliver (composite-jewel cells) we prove the minimal charts pair-barren, exhibit the stratum's genuine pairs (a shear orbit of the germ), prove every scaffold on that orbit impossible at every P-degree, and confine any survivor to non-shear tail deformations with $\deg_W P \geq 4$ and $3 \mid J$. The proofs run on a small amount of new machinery — a graded fat-edge kill, a uniform residue theorem with ratio $-1/(N-1)$ and a coset refinement, a trace identity, and fertility/residue laws for composite cone edges — each of which ships with an exact-arithmetic machine certificate. Byte-frozen copies of the two core theorem documents were hash-committed publicly on July 23, 2026, prior to this note.

1 The mechanism

History. The temptation dates to the first wrong proof: Kraus's 1884 argument [7, 8] failed exactly at controlling ramification at infinity. Vitushkin [6] constructed the honest prototype: a rational map of $\mathbb{C}^2$ with constant Jacobian determinant -2 that is non-injective — legal only because one coordinate carries a pole, on a line. For twenty-seven years the question was whether the pole can be *cancelled* without destroying the constant Jacobian. In July 2026 it was answered in dimension 3: the map

$$F(x,y,z) = \big((1+xy)^3 z + y^2(1+xy)(4+3xy),\\ y + 3x(1+xy)^2 z + 3xy^2(4+3xy),\quad 2x - 3x^2 y - x^3 z\big),$$

announced by Alpöge and derived by the Claude model Fable [1], is a polynomial Keller map ($\det JF \equiv -2$) with an explicit fiber collision; we have verified both facts in exact arithmetic

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(§5). Structurally $F = H \circ G$ where G carries the rational coordinate $w = y + 1/x$ (poles on the hyperplane $\{x = 0\}$) and H is a $3:1$ ramified cubic whose Jacobians cancel to -2 [4, 3, 2]. The conjecture is therefore false for all $n \geq 3$ (and the Dixmier conjecture falls by stable equivalence [5]); dimension 2 remains open.

The plane question. The published lenses on why F works — Tao's multiplication-map digestion [3], the Sawin–Levinson $\mathbb{A}^1$-bundle picture and Gallagher's non-properness account [2] — all use the third dimension as room to hide ramification. Our question is the mechanism's shadow in $\mathbb{C}^2$: can a rational coordinate with poles on a curve Γ be polynomially recombined into a plane Keller map? The pole locus Γ plays the role of the Jelonek non-properness set.

2 Scaffolds and the dichotomy

Definition 1. *Fix a curve $\Gamma \subset \mathbb{C}^2$. A* scaffold over Γ *is a triple (w, c, P) with $w \in \mathbb{C}(x, y)$ having poles exactly on Γ; $c \in \mathbb{C}[x, y]$; and $P \in \mathbb{C}[W, C]$ such that $f := P(w, c) \in \mathbb{C}[x, y]$ (the poles cancel) and $\det \mathrm{J}(f, c) = \kappa \in \mathbb{C}^*$. Write $D := \deg_W P$.*

Over the line $\Gamma = \{x = 0\}$, normalize $w = a/x^d$ with $x \nmid a$ and $c = x\,u$. The chain rule forces the *pair equation*

$$\mathrm{J}(w, c) = \lambda x^m, \qquad \lambda \in \mathbb{C}^*,\ m \in \mathbb{Z},$$

and differentiating $f = P(w, c)$ gives the *master identity* $P_W(w, c) = \mu x^{-m}$ with $\mu = \kappa/\lambda$. Pairs (w, c) satisfying the pair equation are the raw material; scaffolds are pairs that additionally support P. Killing pairs kills scaffolds; several strata contain genuine pairs that die only at the P-layer.

Theorem 2 (Line Dichotomy)**.** *For any line scaffold, $u(0, y)$ is forced to be a nonzero constant γ, and exactly one of the following holds:*

1. *u is constant, and the scaffold is an affine triangular automorphism; or*
2. *u is nonconstant, and $F = (f, c)$ is a Keller map of $\mathbb{C}^2$ that is not injective: the fiber $\{c = 0\} = \{x = 0\} \sqcup \{u = 0\}$ has two components glued by P. Such a scaffold* is *a counterexample to the two-dimensional Jacobian conjecture.*

So the stakes are exact: proving every line scaffold lands in case (1) is precisely proving this mechanism cannot falsify the plane conjecture over a line.

3 The two theorems

Theorem A (Hyperbola Sterility). *Over $\Gamma = \{xy = 1\}$ no scaffold exists at all.*

Theorem B (Line Sterility, modulo one named sliver). *Partition line configurations by the fiber hull of $c - t$ at $x \to 0$ (the lower-left Newton hull from the $-t$ vertex $(0, 0)$ to the top vertex (α, N), $N := \deg_y c$). The partition of Table 1 is complete — the classifier is executable, and a 2,100-shape randomized sweep lands every candidate in exactly one stratum (§5). In every stratum except E4, no scaffold with nonconstant u exists; consequently every scaffold there is an affine triangular automorphism.*

Stratum	Configuration	Closing result
A	single hull edge, open-edge support	discriminant-ring evaluation
B	single edge, no open-edge support	Fat-Edge Kill ($\mathrm{J}(w,c) \notin \mathbb{C}[x, 1/x]$, all m)
C	≥ 2 edges, first edge bare-ramified	Ramified-Edge Kill (same graded proof)
D	≥ 2 edges, $N_1 \geq 2$, multi-supported	divisor-law kill + chart recursion
E0	≥ 3 edges, $N_1 = 1$, simple deeper roots	third-edge undershoot + recursion
E1	two edges, binomial cone edge	$N = 2$: jewel ladder; $N \geq 3$: Uniform Residue
E2	composite cone, repeated root	no-conspiracy telescoping
E3	composite cone, off the fertile locus	Composite-Cone Fertility Lemma
E4	composite cone, *on* the fertile locus	**the sliver** — see below
F	≥ 3 edges, repeated deeper roots	no-conspiracy + trace/character kills

Table 1: The ten strata of the line partition and their closing results.

The sliver, honestly scoped. E4 is nonempty as a *pair* stratum: at the first fertile family ($(N, r) = (3, 6)$, interior cone coefficient $t = \pm\sqrt{3}$, pole depths $k \equiv 1 \pmod 3$; machine-confirmed $k \in \{1, 4, 7\}$) the local germ exists in closed form, $w_{\mathrm{germ}} = -\lambda(x^{-1} + \sqrt{3}\, x^2 Y)$ at $k = 1$, and the shear $S\colon Y \mapsto Y + \gamma x$ carries it to a genuine pair family — the composite analog of the binomial shear orbit. We prove:

1. the residue ratios at the cone's punctures are primitive cube roots of unity, so the residue clash kills every scaffold with $3 \nmid J$;
2. the character/divisor bound confines survivors to $D \geq 4$;
3. minimal charts (no tails beyond the forced γx) are barren at the pair level: the bucket march terminates with an unshieldable cokernel obstruction at the germ's first γ-rung, with core $(k-2)\, t\gamma\lambda$, at $k = 1, 4, 7$ and both signs of t;
4. *the entire shear locus is scaffold-dead, for every realization σ and every P-degree*: the master identity decouples over the germ pair's weight grading; functional independence ($\mathrm{J}(w, c) \neq 0$) forces every non-target weight line of P_W to vanish as a polynomial; a nonzero μ then forces the single coefficient $a_{\sigma-1,0} \neq 0$, whose contribution to the sheared $f \circ S$ is an uncancellable leading pole $a_{\sigma-1,0}(-\lambda)^{\sigma-1} x^{-(\sigma-1)}$ along $Y = 0$.

What remains open of the entire program: non-shear tail deformations of these pairs — nonempty at the pair level (an exact window candidate exists) and confined to $D \geq 4$ with $3 \mid J$ — which we expect to fall to the ξ-ψ-calculus, the cone ODE being first order in ξ. Every other stratum is closed.

Corollary. *The mechanism that falsifies the Jacobian conjecture in dimension 3 cannot produce a plane counterexample over the standard hyperbola, nor over a line outside stratum E4 — and inside E4 only via non-shear tail deformations with $D \geq 4$ and $3 \mid J$.*

Not claimed: other pole curves (general $\mathbb{C}^*$-embeddings, higher genus, reducible Γ), non-triangular compositions, or the two-dimensional conjecture itself.

4 The new machinery (statements)

Theorem 3 (Fat-Edge Kill / Staircase Sterility)**.** *Suppose the support of c satisfies the strict collapse hypothesis for the grading $\omega(i,j) = Ni - \alpha j$: $\omega > 0$ off the top vertex $b\,x^\alpha y^N$. Then for every line candidate w, $\mathrm{J}(w,c) \notin \mathbb{C}[x, 1/x]$ — no pair, for any $m \in \mathbb{Z}$, zero included. The proof is one page: the top-vertex bracket $L_0 = \mathrm{J}(\,\cdot\,, bx^\alpha y^N)$ is diagonal in the monomial basis with eigenvalue $b\omega$ and shifts y-degree by $N-1$. A verbatim ω_1-graded variant (Theorem′) kills bare-ramified first edges.*

Theorem 4 (Uniform Residue)**.** *In every fertile binomial cell (N,k,r) — fertility being exactly $r \mid Nk$ and $r \nmid k$ — the normalized germ residue ratio at every puncture is $-1/(N-1)$, independent of k, r, and all tails. The induced clash forces $(N-1)^J = 1$ against the master identity, killing all scaffolds in stratum E1 with $N \geq 3$; $N = 2$ (ratio -1, silent on odd D) falls to the jewel ladder and parity-pole argument instead. At $d = \gcd(N-1, r) > 1$ the germ decomposes over ζ_d-cosets; the operator preserves cosets and a kernel-exclusion argument closes the refinement (witnessed at $(3,2,6)$).*

Theorem 5 (Composite-cone laws)**.** *With interior cone-edge support (cone polynomial $E(\xi)$, $d = \gcd(N-1, r) > 1$): the germ recursion acquires a termination condition $P_k(t) = 0$, an explicit polynomial per k (at $(3,6)$: $P_1 \propto t^2 - 3$, $P_2 \propto t(2t^2 - 9)$), off whose zero locus the cone is germ-barren (closing E3); on it, the per-root residue ratio is*

$$\frac{\varphi(\rho)}{\varphi(0)} = \frac{d}{(N-1)\,\rho\,E'(\rho)},$$

whose values at the first fertile family are the primitive cube roots of unity above. The binomial constant $-1/(N-1)$ is the special case $E = 1 + \xi^d$ evaluated at its roots.

Alongside these run a trace identity (sum of residues over each cone fiber's punctures vanishes), the divisor-law/no-conspiracy telescoping for multi-edge staircases, and the sharpness objects: the $N = 2$ *jewel* pairs (e.g. $w^2 - 4c = x^{-2}$ with involution $(x,y) \mapsto (-x,\, y + 1/x^5)$), the shear orbits, and the composite-jewel germs. All die at the P-layer or in the confined sliver — none assembles into a counterexample.

5 Machine certificates and priority

All results are backed by exact-arithmetic sympy witnesses (no floats, no Monte Carlo in any proof-bearing step): witness suites totalling over 3,000 individually asserted checks, including the partition classifier (2,100 randomized shapes plus a named zoo, every shape in exactly one stratum), the bucket-march records for the composite-jewel kills, and the corrected sheared-row P-layer sweeps. Six independent adversarial model audits (two GPT-family, GPT-Pro, Gemini, Claude-family, and a codex delta-audit of the newest results) ran against the documents; every confirmed finding is folded and changelogged, including one repair (the gcd-coset gap) independently re-derived by two models coefficient-for-coefficient, and the delta-audit's contribution of the all-(σ, D) shear-locus argument. The witness repository accompanies this note.

Priority: byte-frozen copies of the two core theorem documents were committed publicly by SHA-256 on July 23, 2026 (X/Twitter), prior to this note:

```
a40f311c2ee8aaabd3dada15b05dca81ef9dda540dcda4bb18c1b98352548dfc
58633ec91872c9746f3d6192d38dce0c106f88d94d0637dcf5f5815f3b06dc78
```

This note reveals the committed content; the frozen files ship with the witness repository.

6 Relation to prior and concurrent work

Gwoździewicz [9] proved a Keller map injective on one line is an automorphism — a foil: our theorems constrain the *mechanism*, not the map's restriction. The Newton-polygon school (García Vicente–González Villa et al., Makar-Limanov) supplies the polygon vocabulary; our fiber hull, gradings, and residue calculus run at the pair-equation level the polygon alone does not see, and the school's own papers state that boundary. Orevkov's analytic realizations of Vitushkin's covering [10] are the closest classical geometry to the pole-hiding lens. Concurrently with this note, Shaska [11] proved every $\mathbb{G}_m$-equivariant (graded) Keller map of $\mathbb{C}^2$ is an automorphism; the class boundary is exactly our subject — scaffolds are not graded, and the tails that break equivariance are where jewels, shears, and the sliver live. On the fiber-topology side we use Abhyankar–Moh–Suzuki, Zaidenberg–Lin, and Kaliman's classification in the $D = 1$ closure, and Masuda–Miyanishi sits adjacent to the jewel geometry. To our knowledge the sterility theorems, the fat-edge kill, the uniform residue law with its coset refinement, the trace identity, and the composite-cone fertility/residue laws are new.

7 Methods and contributions

The mathematics in this program was derived by AI systems under human direction, in the lineage of the dimension-3 counterexample itself [1] and Gallagher's follow-up [2]. Primary derivations and all machine certificates: Claude (Fable 5, Anthropic), across a multi-session campaign with persistent memory and a theorem ledger. Independent adversarial audits and parallel derivations: GPT-5.6-sol (OpenAI), whose $(3, 2, 6)$ coset analysis matched the primary derivation coefficient-for-coefficient, and whose delta-audit of the composite-jewel results corrected one reduction step and contributed the core of the all-(σ, D) shear-locus argument; additional audit rounds by GPT-Pro, Gemini, and Claude-family reviewers. Direction, verification infrastructure, cross-model transport, and the decision to work the plane question: Liam Corrigan. Every theorem-bearing claim traces to an exact-arithmetic witness script; audit history and rebuttals are preserved in the working documents.

References

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[6] A. G. Vitushkin, *On polynomial mappings of* $\mathbb{C}^2$, Math. Notes **66**:2 (1999), 245–249. [Springer PDF pull pending before submission; formula currently cited via verified paraphrase.]

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